Intrusion Prevention System of Automotive network CAN bus

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***Abstract*— Automotive networks based on the Controller Area Network (CAN) bus (ISO 11898) family of protocols have been shown to be vulnerable to exploits by hackers who are outside the vehicle. These networks can be compromised in a manner which could jeopardize vehicle occupants. One well-publicized exploit led to a costly automotive recall that affected more than a million vehicles. Other exploits can allow criminals to steal cars without physically breaking into them. While no fatalities have yet occurred, hackers could trigger an event that led to accident involving serious injury or even death. The CAN bus connects electronic control units (ECUs), some of which are required for safety and emissions systems such as the anti-lock braking and fuel injection systems. In addition to controlling required functions, other ECUs provide consumer-oriented features such as infotainment and lighting. Whether factory installed or aftermarket add-ons, each ECU introduces attack vectors into the overall automotive network. This research focuses on securing these vehicle networks, specifically the CAN bus. This paper will analyze the current vulnerabilities and describe our design for a real-time intrusion prevention system (IPS) that neutralizes attacks by actively monitoring the CAN bus and eliminating malicious messages.**

***Index Terms*— Cyber security; IDS; Automatic Threat Recognition.**

I. Introduction

Modern vehicles rely on communications buses that were designed for closed systems in the 1980's. The protocol is robust in noisy environments and is ideal for communications in the modern automobile. However, security is not always considered when a protocol is first designed and implemented. The CAN bus was designed as a closed system so protection against outside attack was not an important consideration. As the technology in vehicles has evolved, the current implementations are no longer closed systems which opens the possibility for attack.

The vulnerabilities of the CAN bus include shorting, applying a high voltage, denial of service, and replay attacks. This paper describes a system that mitigates the replay attack and limited denial of service attacks.

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A. Vehicle wiring

The automotive wiring harness has undergone vast changes as electronics were introduced into vehicles. Manufacturers of automobiles improved the capabilities of their models as technology progressed and mechanical systems were replaced by cyber-physical systems. The Antilock Braking System, airbags, electronic locks, transmission control, and active suspension are a few examples of comfort and safety integrated into vehicles as technology progressed. Demand for greater pollution control and fuel efficiency led to legislation such as the Clean Air Act [1], which mandated manufacturers to design vehicles with reduced exhaust emissions. The electronics to control the harmful emissions added to the complexity of the wiring harness. The increased number of connections and point to point wiring increased material cost, production time, and troubleshooting time during manufacturing and servicing. The reliability of the numerous connectors exchanging data created a noisy environment which led to an increase in maintenance, fault diagnosis, and repair times.

A car alarm or other aftermarket electronic device requires an extensive amount of wiring for functionality. The installation of the aftermarket electronics requires someone installing the equipment to have extensive knowledge of the wiring harness of the vehicles. Each manufacturer (Chevrolet, Ford, Dodge, etc) uses different wiring schemes based on the make and model of the vehicle. Figure 1 shows an example of how an aftermarket remote starter and alarm system must be wired to each device it will monitor.

As the complexity of the control system and the number of wires and connectors increased, manufacturers sought a cost-effective real-time vehicle bus system for sensor, controller, and

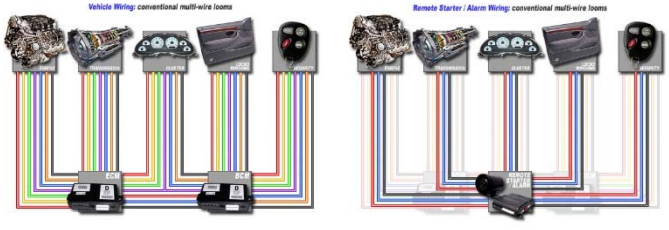


Figure 1 Wiring Harness before CAN bus [4].

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actuator communication. Robert Bosch GmbH introduced the serial CAN bus protocol at the Society of Automotive Engineers (SAE) congress and it is now an international standard supported by ISO [2].

After the introduction of the CAN bus, the wiring harness in the automobile changes from point-to-point connections into a fieldbus with electronic controllers receiving broadcast messages from nodes. With bus protocols, vehicular wiring can be reduced by 75 percent [3]. Figure 2 shows the reduced wiring requirement for a remote starter and alarm system on a CAN bus-equipped vehicle. The legacy system simply connects to a gateway node on the CAN bus.

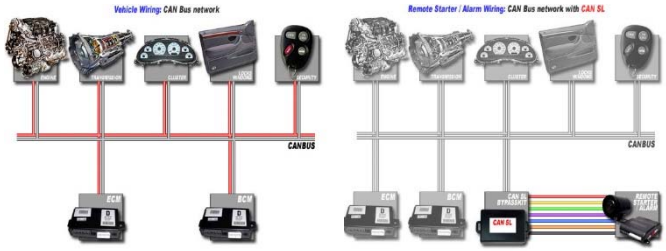


Figure 2: Wiring Harness after CAN bus [4].

To continue to enhance comfort and safety, additional control modules have been added to vehicles, which caused other networks with different protocols to be added to vehicles. Modern automotive networks use a combination of CAN buses for mainstream powertrain and body communications, LIN for low-cost applications primarily in body electronics, FlexRay bus for high-speed synchronized data communications in advanced systems such as active suspension, Media Orientated Systems Transport (MOST) for multimedia communications, and other proprietary networks that are manufacturer specific such as General Motors Local Area Network (GMLAN). Each of these networks have different requirements and stringent specifications from the SAE.

B. *Vehicle Networks*

1) *Controller Area Network (CAN bus)*

The CAN bus is a multi-master, asynchronous, message broadcast system designed to operate in a noisy environment. The CAN protocol is internationally standardized in ISO 11898-1 and 11898-2. The CAN bus comprises the data link layer and physical layers of the 7-layer ISO/OSI reference model. CAN is based on the "broadcast communication mechanism," which is based on a message-oriented transmission protocol. It defines message contents rather than nodes and node addresses. Every message has a message identifier and an arbitration identifier, which is unique within the whole network since it defines content and also the priority of the message. CAN bus arbitration enforces the Carrier Sense Multiple Access / Collision Avoidance (CSMA/CA) which prevents collisions on the bus when several nodes compete for access to the bus, also known as bus arbitration [4]. The various CAN buses operate at different data rates depending on the cost, performance, safety, and node requirements. The normal CAN bus data rates are 125 kbps, 250 kbps, 500 kbps, and 1 Mbps, but other data rates are used for manufacturer-specific implementations.

2) *General Motors Local Area Network (GMLAN)*

GM Local Area Network (GMLAN) is General Motor's implementation of the CAN bus on GM vehicles. The GMLAN supports three separate buses: low-speed (33.33 kilobits per second), mid-speed (95.2 kilobits) and high-speed (500 kilobits) [3].

3) Flexray, Most, and LIN

These networks also provide critical communications in the vehicle, but are not the focus of this paper.

C. *Vehicle Control Modules*

The Electronic Control Module (ECM) or the Electronic Control Unit (ECU) in the modern automobile was designed to improve functionality by reducing exhaust emissions and increasing fuel economy [8]. The ECU is an embedded microprocessor designed to control a subset of the functionality of the vehicle. Since the inception of the ECM / ECU, there have been several control modules designed to increase comfort and safety. The ECU's are distributed throughout the vehicle controlling the various subsystems. Some of subsystems requiring control modules include: the Transmission Control Unit (TCU), the Power Train Module (PCM), the Body Control Module (BCM), the Ignition Control Module (ICM), and the Cruise Control Module (CCM). Each of these modules are interconnected via a gateway module connected to the various automotive buses. The various ECU's receive digital and analog signals. ECU's process input from sensors using look-up tables. For example, if the coolant temperature sensor sends an analog reading to the ECU which is converted to a digital signal by an analog-to-digital (A/D) convertor and represents a temperature of 180 degrees; the ECU will use its lookup table to determine if it is necessary to activate the coolant fan.

D. *Diagnostics*

On-Board Diagnostics, or "OBD," is a computer-based system mandated by the 1990 Amendments to the Clean Air act to be installed into all light-duty vehicles and trucks sold in the US since 1996. OBD systems are designed to monitor the performance of some of an engine's major components including those responsible for controlling emissions. The EPA website discusses the benefits of the OBD port and how the port can be used to show the emissions and troubleshooting information [9].

The OBD-II specification in Figure 3 specifies standardized hardware interface—the female 16-pin (2x8) J1962 connector. Unlike the OBD-I connector, which was sometimes found under the hood of the vehicle, the OBD-II connector is required to be "located in the passenger compartment in the area bounded by the driver's end of the instrument panel to 300mm beyond the vehicle centerline, attached to the instrument panel, and accessible from the driver's seat [9]."

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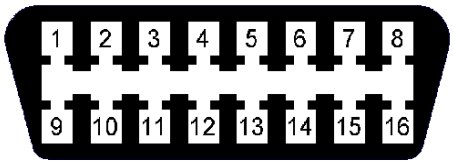


Figure 3 OBD-II Connector [10].

Although originally designed for tasks such as maintenance, troubleshooting and emissions certification the OBD-II port is now being used for many additional purposes. Some examples include insurance companies monitoring driving habits with devices such as Progressive's "Snapshot *[11]*." Another is "Hum" developed by Verizon which is a system that provides a way to access vehicle data as well as contact help (similar to OnStar). In both systems, there is a small device that plugs into the OBD-II port and also connects to a cell phone network. These systems open the "closed" CAN bus network to outside traffic and could potentially be used as attack vectors.

The devices which monitor the vehicle data generally plug directly into the OBD-II port. Connecting a computer to the OBD-II port often requires an adapter cable which maps the OBD-II pin to a DB-9 connector. The recommended pinouts from CAN in Automation (CiA), are shown in Table I. Companies like Peak systems have adopted the CiA recommendation and others like SparkFun use the specification of SK Pang Electronics who created the Arduino CAN bus shield used in this research.

TABLE I. OBD-II PINOUTS DB-9 [20]

Pin	Signal	Description
1	-	Reserved
2	CAN_L	CAN_L bus line (dominant low)
3	CAN_GND	CAN Ground
4	-	Reserved
5	(CAN_SHLD)	Optional CAN Shield
6	(GND)	Optional CAN Ground
7	CAN_H	CAN_H bus line (dominant high)
8	-	Reserved (error line)
9	(CAN_V+)	Optional CAN external positive supply (dedicated for supply of transceiver and optocouplers, if galvanic isolation of the bus nodes applies)

E. The Connected Vehicle

As consumers demand enhanced capabilities, such as integration of their smart devices to allow for content streaming through the automobile entertainment system, and governments continue to enforce tighter emission and fuel economy standards, manufacturers will continue adding functionality to meet the demand. Vehicles equipped with GM OnStar, Toyota SoS, or other manufacturer-specific assistance systems connect external communication sources to the closed automobile system. Other external sources already available to the consumer include WiFi, Bluetooth, USB, and 3G/4G/LTE. The U.S. Department of Transportation (DoT) Research and Innovating Technology Administration (RITA) has been working since 2009 to establish policies, procedures, and safety guidelines for short-range vehicle communications. The goal is to deliver data and create a dynamic data exchange from vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), and vehicle-to-mobile device (V2D). The National Highway Traffic Safety Administration's (NHTSA's) current research indicates that the V2V system will use a Public Key Infrastructure (PKI) to authenticate messages [21].

II. RELATED WORK

A. Automotive Security Road Map

Larsen and Nilsson lay out a roadmap for the security of the modern automobile [12]. They describe a vehicle as a Cyber-Physical System (CPS) which relies on external inputs to accuate physical devices such as the brakes, accelleration, or steering. The fundemental limitations of the CPS are operation in a real-time environment, limited computation power, and the difference between vehicle network traffic patterns and those found on traditional IP networks. They point out the following research challenges:

1) *Cyber attacks simulating the physical world*
2) *Authentication of incoming data to the vehicle.*
3) *Replay attacks*
4) *Non-repudiation of messages*
5) *Wireless connection to the internal network*
6) *Firewalls to prevent unauthorized access (Access control list)*
7) *Logging and detection mechanism to track attackers*
8) *Cryptographic security on wireless connections*
9) *Protection of the physical OBD-II port*

The paper goes on to lay out the roadmap for 5, 10, 20 years according to the defense-in-depth concept [13]. The goal for the roadmap is to emplace security defences in the five defense layers prevention, detection, deflection, countermeasures, and recovery In the five year map, they suggest prevention and detection of attacks should be the main focus including communication between ECU's to prevent injections of messages.

B. Vulnerability Assessment

Koscher et al. [14] and Checkoway et al. [15] are the seminal papers in the field of automotive vulnerability assessment. In the first paper, they evaluated two identical vehicles on a closed test track, on blocks, and on the test bench by removing specific components. A custom program, CARSHARK, was created to take advantage of the vulnerabilities found. Vulnerabilities were found during monitoring of the CAN bus while plugging in the various components to determine the messages associated with the component. After determination of the arbitration identifier, introducing packets containing a valid identifier but random data caused actuation in the cyber-physical vehicle. Elimination of the replay attack would prevent this fuzzing capability and the associated responses which led to targeting of specific subsystems. Additionally the authors were able to

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inject malicious code into the telematics system allowing them to send custom commands on the CAN bus.

Their second paper focused on the attack surfaces. While eliminating the replay attack would not foil all of the attack vectors, it would severely diminish the effectiveness of most of the vulnerabilities. Table II shows the potential vulnerabilities and where replay can be utilized to control the various subsystems.

Table II. Attack Types[15]

Type of attack	Injection medium	Replay used
Direct	OBD-II	Yes
Indirect	Pass-thru on OBD-II	Yes / No
Direct	Recharging cable electric vehicle	Yes
Direct	CD / USB / IPOD	Yes
Indirect	Bluetooth/Keyless entry/TPMS/RFID/WiFi	Yes
Indirect	GPS/Sat radio/HD radio/RDS/TMC	Unknown

Reprogramming the ECU's with custom firmware to give the intruder access to the internal network was the primary means in controlling the vehicles they tested. Their work clearly demonstrated the ability to exploit the vehicle remotely and control the ECU's by injecting malicious code. They were able to get remote code execution on a telematics unit of a vehicle by exploiting a vulnerability in the Bluetooth stack of an ECU and separately compromising a cellular modem. The cyber-physical vehicle could then be controlled by executing commands resulting in behaviors which were not intended when designed.

Miller and Valasek [16] extended the work of Kocher et al. by implementing new attacks and building a framework that will allow the construction of tools for automotive systems, this would allowing researchers to monitor and demonstrate vehicle vulnerabilities. They successfully dissected a Ford Escape and Toyota Prius showing they could control almost all functionality in the vehicles. The research team were able to demonstrate their attacks while the car was being driven. Histograms of the CAN bus demonstrated the cyclic nature of some messages and by detecting diagnostic packages during normal operation, anomalies can be detected. They concluded that automobiles are designed with safety in mind but one cannot have safety without security in mind. Their goal was to show the need for safer ECU's in a hostile CAN bus network and the ability to detect and stop attacks on the vehicle network.

Miller and Valasek [17] additionally conducted a survey of numerous modern vehicles looking at how the remote attack surfaces have evolved over time. They quantify the attack surfaces of the cyber-physical features installed in the vehicle and then recommend some defensive strategies such as an Intrusion Detection System (IDS) to prevent the attacks on the CAN bus. The research defined the anatomy of remote attacks in three stages.

1) Remotely gaining access to an internal automotive network.

2) Communicate with safety critical ECUs.

3) Make the target ECU behave in some way that compromises vehicle safety.

After discussing the hackability of each model they propose some security measures which can help secure the automotive systems.

a) Securing remote endpoints

b) CAN injection Mitigations

c) Message Cryptography

d) Network Architecture

e) Attack Detection

They additionally built a device[17] which plugs into the OBD-II Port, learns the traffic patterns and then detects anomalies. However, the device shorts the CAN bus when an anomaly is found and this shuts down the bus to all normal traffic. They conclude by suggesting a defense in depth strategy to secure the system overall.

III. CONNECTING TO THE CAN BUS

A. Connectivity

Connection to the CAN bus varies but is rather straightforward. The bus is generally implemented as a twisted pair of wires terminated at both ends by 120 Ω resistors. For this research, the connectivity was implemented as shown in Figure 4. The USB2CAN may be replaced with a BT2CAN or a WiFi2CAN enabling wireless connectivity to the vehicle bus. An adaptor cable is required to connect the DB-9 USB2CAN to the 16 pin ODB-II port on the automobile.

Once the connectivity is complete, the bitrate of the bus needs to be configured. Some normal bus bitrates are 125 kbps, 250 kbps, 500 kbps, and 1 Mbps, but for GMLAN bitrates are as low as 33.333 kbps and require a different transceiver for connection to the bus. The High Speed (HS) bus in the three test vehicles were all set to 500 kbps.

B. Safety

While conducting our research, the following safety considerations were taken on the vehicle which included removing the fuse for the air bag system to prevent unintentional deployment, chocking the wheels to prevent movement, and having an occupant in the driver's seat to engage brakes and perform tasks from the list. With the exception of recording the starting of the engine and activation of the rear camera, the testing was done with the engine off and the ignition key in the "run" position. Extended testing required a battery charger to prevent the battery from discharging below the usable voltage.

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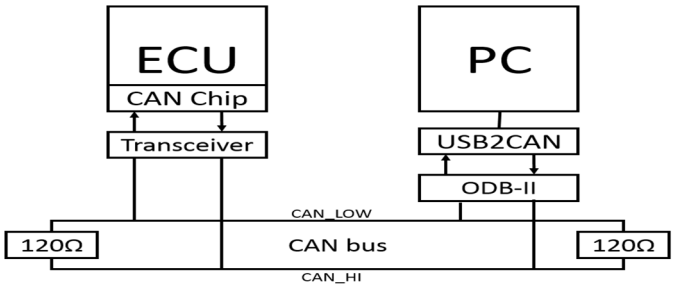


Figure 4 Two node CAN bus network

C. CAN bus Timing

The CAN bus is a pseudo-synchronous communications bus which supports bit rates from 1 kbps to 1Mbps. Each node on the bus contains an oscillator responsible for controlling timing. Due to slight variations in the frequencies due to voltage or temperature changes, there is a CAN bit synchronization algorithm [18] [19] which allows for resynchronization of the bit stream. The nominal bit timing shown in Figure 5 is comprised of four segments which are the synchronization, propagation, phase buffer 1, and phase buffer 2 segments. The synchronization segment is normally one time quantum (t_q) and the propagation segment normally compensates for the time delay from the two farthest nodes on the bus. Proper engineering of the bus results in a common bit rate throughout the CAN bus segment.

Similar to Ethernet, CAN bus segments can be joined together using repeaters, bridges or gateways. The repeater connects the two segments while logically being one bus and introduces a small delay between the segments.

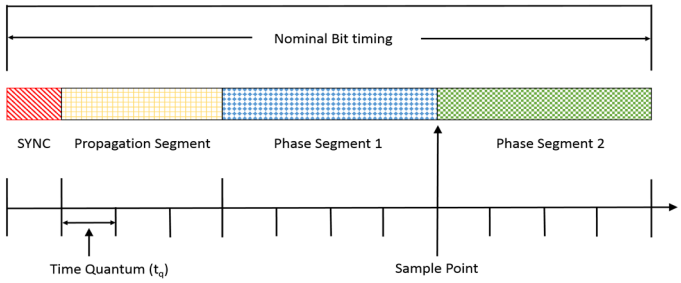


Figure 5 CAN bus Bit Timing

The repeater transfers all electrical signals and arbitration identifiers are common on both segments. The bridge connects two segments and has unique arbitration identifiers on each physical segment. The bridge can forward non-local messages between segments and operates at the data-link layer. The gateway functions as a translator and operates at the application layer. The gateway can forward messages from buses with different higher-layer protocols.

IV. Intrusion Detection System

A. Concept

The intruder or hacker, shown in Figure 6 can be attached to the vehicle bus via the OBD-II port, wirelessly, or hardwired. Figure 7 shows an example implementation of control code to detect the replay attack. Each node would require an implementation of the control code. A separate anomaly detector can be configured to capture timing events or Denial of Service as suggested by [14] [15], log invalid arbitration identifiers from the control code in Figure 8, and log replay attacks detected by the control code in Figure 7.

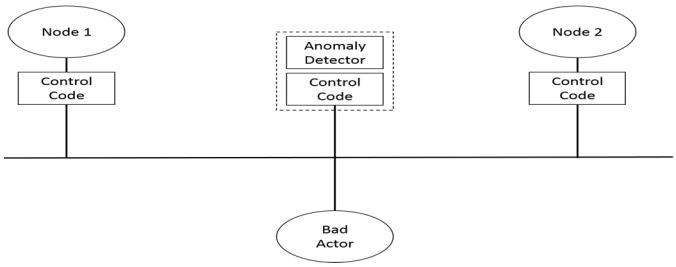


Figure 6 Replay Elimination and Anomaly Detector

1) Valid message

A valid message in the automotive bus is one which was designed by the manufacturer and encoded into the various ECU's on the CAN bus logical segment. In the current implementation on the CAN bus protocol, hackers can capture the valid messages and use the messages to make the vehicle preform actions at inappropriate times causing drivers to be distracted.

2) Replay messages

A replayed message is a message which has been captured from a CAN bus segment or is already known. The message which is comprised of the arbitration identifier and data, can then be used to fuzz the bus by using the identifier and either sequencing through the data or randomly choosing data bits. The algorithm in Figure 7 listens to the idle bus and once the SOF in the arbitration occurs, it matches the arbitration identifier to the ones preprogramed in the ECU. If there is an arbitration identifier match and the ECU is not transmitting then the message is from an unknown node indicating a replay attack is occurring. The ECU would then send a message to the anomaly detector alerting that a replay attack has taken place. Alternatively, the manufacturer could predefine a message which the anomaly detector would automatically register as an anomaly. The node will have a white list of all known arbitration identifiers which belong to the node. The node will have non-repudiation as it will notify about invalid messages which did not originate from the authoritative node. In the intrusion protection system the node can cause the spoofed message to error out.

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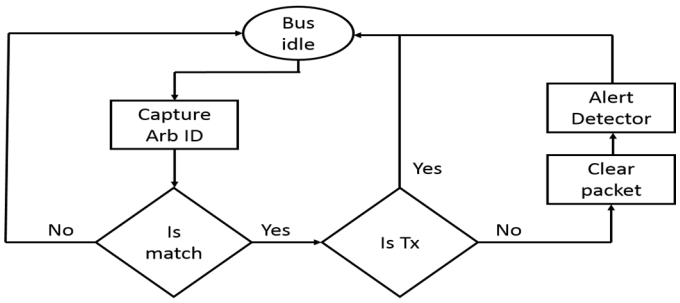


Figure 7 Control Code for Replay Attacks

3) Invalid messages

An invalid message is one which has an arbitration identifier which is not associated with the ECU's on the CAN bus segment. Each logical segment would require a device which could be implemented in a gateway which would monitor the bus traffic with the algorithm in Fig 8. The device would monitor all traffic and compare the arbitration identifiers to the preprogrammed valid identifiers from the manufactures. Like the control code implemented in each ECU, the anomaly detectors control code would listen for arbitration identifiers which are not part of the logic CAN bus segment. The control code would notify the IDS of an unknown arbitration identifier.

The IDS would detect the replay of valid arbitration identifiers through monitoring timing events, monitoring known messages being transmitted from unknown nodes, and detecting unknown arbitration identifiers which do not belong to the bus.

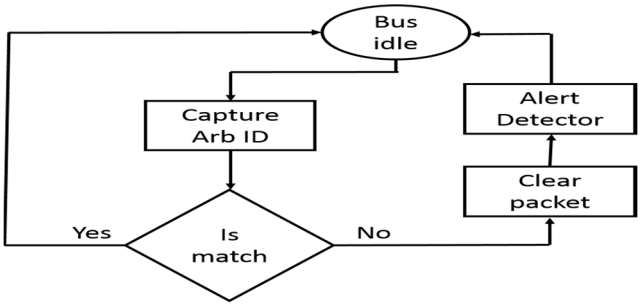


Figure 8 Control Code for Invalid Arbitration Identifiers

B. Simulation

Implementing the control code requires a FPGA or ASIC to extend the capability of the base system. For this research we have chosen a DE2-Nano FPGA from Altera to handle the control code. Table III shows the list of components used to implement the control code and CAN bus nodes.

TABLE III List of node components

Device	Purpose	per node
Arduino UNO	Microcontroller for decision making	1
CAN bus shield	Receives/Transmits (CAN)	1
DFRobot LCD	I/O to demonstrate functionality	1
Altera DE-NANO FPGA	Detects messages on node	1
Bi-directional level shifter	3.3V to 5V Arduino to FPGA	1

The Altera software package includes the ModelSim simulation package which can be used to run simulations on the control code programmed in VDHL. The scenario for this example includes the traction control system and the windshield wipers. The traction control was given the arbitration identifier of "2F7" and the wipers were given "4AB." Figure 9 shows the results of the simulation in ModelSim. The different lines from top to bottom are:

1) *The RX line – displays bit values of input to the node*
2) *The TX line – displays bits being transmitted from node*
3) *Current TX – state machine tracking traction control*
4) *TX match – indicator that current node is transmitting*
5) *Current state – 2us timing*
6) *TC match – indicates a traction arbitration identifier*
7) *Current WP – state machine tracking wiper control*
8) *WP match – indicates a wiper arbitration identifier*

The first box in the waveform in Fig 9 is the "2F7" indicating a traction control message is being received on the RX line. However, the TX match line is indicating the node is not transmitting. These two conditions combined show that the message is being replayed from an unknown node. This triggers the TC match as shown in Fig 9 to be driven logically high to alert the IDS of a hacking event.

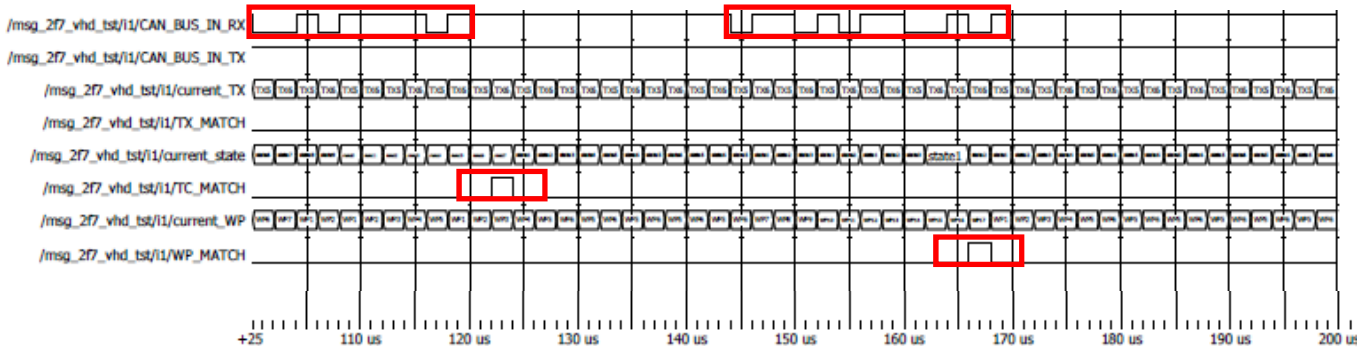


Figure 9 Altera Modelsim simulation

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The second box on the RX line is the arbitration identifier "4AB" for a wiper message. Likewise the TX match line remains at a logical low indicating the node is not transmitting. Again demonstrating a message which is coming from an unknown node. The control code drives the WP match logically high to designate a replay attack to the IDS.

C. Hardware

Figure 10 shows the assembled prototype with the top three nodes containing the parts described in Table III. The devices with the LCD shields represent typical ECUs and the devices with the LEDs are the FPGA and level shifter which contains the control code that monitors the CAN bus. The node in the lower left of Fig 10 is the logger which passively monitors the CAN bus for alerts from the control nodes. The final node in the lower right emulates the CAN messages from the vehicle broadcasting the RPM, vehicle speed, and coolant temperature.

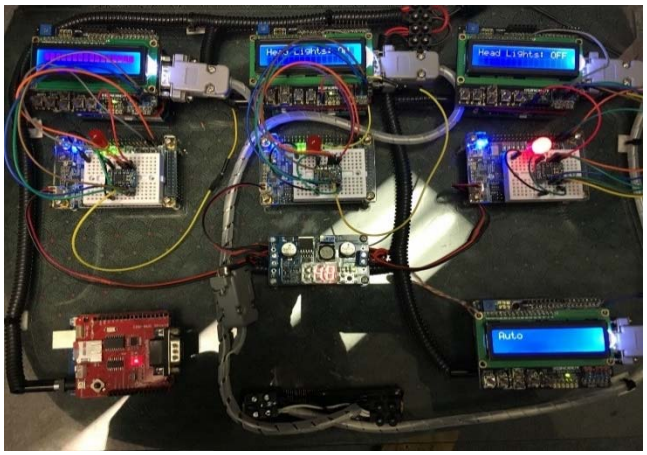


Figure 10 Complete IDS prototype

Capturing the logic on the lines is done by using a Saleae logic probe which can decode multiple protocols, and for our research the CAN bus messages. Figure 11 shows the implementation of the traction control detection.

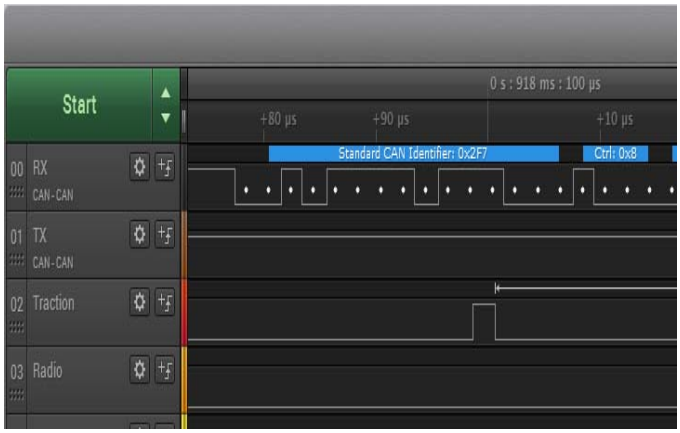


Figure 11 Traction control identifier detected

Figure 12 shows the authoritative node transmitting the message. The node receives it on the RX line with a small propagation delay. Since the message originated from the correct node the traction line is not driven high indicating the message is valid.

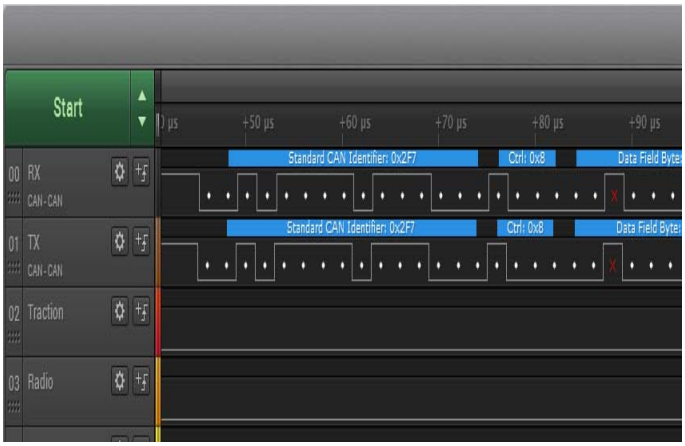


Figure 12 Traction control identifier transmitted from node

D. Intrusion Prevention System

Extending the IDS with an additional CAN transceiver driven from a line from the FPGA, the system can be converted into an IPS. When the traction control replay signal is detected in figure 11, the FPGA will send several dominant bits which will cause an error on the bus and the current CAN message will be invalid. In addition, if the attacker node is using standard CAN chips, they will register the error frames and shut down when the error counter exceeds the threshold. This IPS should prevent rogue nodes from shutting down the CAN network while trying to create a DoS with valid messages.

IV. CONCLUSION

In this paper we have presented a survey of the current research related to vulnerability assessment in the modern automobile's CAN bus, and algorithms to detect replay attacks of both valid and invalid arbitration identifiers. Implementation of a defense in depth approach on the vehicle communication bus should include security on the base network, ECU's, external interfaces, and aftermarket components. Detection of the replay attack will ensure that invalid messages that occur on the logical CAN bus segment will cause an alert to occur up to the point at which a DoS occurs because of an overloaded bus. Manufacturers need to continuously integrate security measures with each new device added to the modern vehicle. It has been well proven that the modern automobile is susceptible to being hacked [12-17]. Ultimately, the CAN bus protocol as designed, due to the broadcast nature, is susceptible to message injects from nodes which have been hacked or attached to the original bus. These nodes can be utilized to passively listen or transmit messages on the CAN bus causing a CPS response. The network changes where each node knows the messages

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assigned to its node. While modification of the CAN bus protocol is unlikely, adding an additional chip (ASIC or FPGA) with the protection algorithm to supplement the CAN bus chip is a way to alert about replay attacks, thus decreasing the motivation to hack the vehicle. The vehicle would not respond to replay attacks as each critical node would be protected with a white list providing non-repudiation at the node level.

Future work

The intrusion detection system will work well for alerting about events occurring on the CAN bus. However, integrating the IDS with an OnStar like service could give manufacturers insight on vehicles being hacked or malfunctioning. Implementing this control code on an actual automotive platform using the organic ECU's would demonstrate the capability of the IDS and provide a platform to capture performance data.

REFERENCES

[1] U.S. Government, "US Environmental Protection Agency - Clean Air Act," 6 January 2016. [Online]. Available: https://www.epa.gov/clean-air-act-overview/title-ii-emission-standards-moving-sources#iia. [Accessed 2 June 2016].

[2] ISO, "Road vehicles -- Controller area network (CAN) -- Part 1: Data link layer and physical signalling.," 2015. [Online]. Available: http://www.iso.org/iso/home/store/catalogue_ics/catalogue_detail_ics.htm?csnumber=63648.. [Accessed 2 June 2016].

[3] AC Delco, "CAN Communication," *TechConnect,* vol. 16, no. 1, pp. 1-3, Jan/Feb 2009.

[4] D. Satkunas, "What is CAN bus?," [Online]. Available: canbuskit.com/what.php. [Accessed 5 March 2013].

[5] ISO, "Road vehicles -- Local Interconnect Network (LIN) -- Part 7: Electrical Physical Layer (EPL) conformance test specification," 10 December 2015. [Online]. Available: http://www.iso.org/iso/catalogue_detail.htm?csnumber=61229. [Accessed 2 June 2016].

[6] ISO, "Road Vehicles - FlexRay communications system," 2013. [Online]. Available: http://www.iso.org/iso/home/search.htm?qt=17458&published=on&active_tab=standards&sort_by=rel. [Accessed 2 June 2016].

[7] MOST Worldwide, "Welcome to the MOST Cooperation," 2016. [Online]. Available: http://www.mostcooperation.com/. [Accessed 2 June 2016].

[8] M. Richter, "Understanding the ECU: What it does and how it works," *MC2 Magazine,* pp. 62-65, Summer 2006.

[9] United States Environmental Protection Agency, "On-Board Diagnostics (OBD)," 2016. [Online]. Available: https://www3.epa.gov/obd/. [Accessed 2 June 2016].

[10] B&B Electronics, "The OBD-II Home Page," 2011. [Online]. Available: http://www.obdii.com/connector.html. [Accessed 2 June 2016].

[11] Progressive Insurance Company, "Snapshot: The fair way to pay for car insurance," 2016. [Online]. Available: https://www.progressive.com/auto/snapshot/. [Accessed 2 June 2016].

[12] U. E. L. a. D. K. Nilsson, "Securing vehicles against cyber attacks," in *CSIIRW '08: Proceedings of the 4th annual workshop on Cyber security and information intelligence research*, New York, 2008.

[13] National Security Agency, "Defense in Depth," 12 March 2010. [Online]. Available: https://www.iad.gov/iad/library/reports/defense-in-depth.cfm. [Accessed 2 June 2016].

[14] A. C. F. R. S. P. T. K. S. C. D. M. B. K. D. A. H. S. a. S. S. K[3] K. Koscher, "Experimental security analysis of a modern automobile," in *IEEE Symposium on Security and Privacy*, 2010.

[15] S. M. D. K. B. A. D. S. H. S. S. K. K. C. A. R. F. K. T. e. a. Checkoway, "Comprehensive experimental analyses of automotive attack surfaces," in *Proc. 20th USENIX Security*, San Francisco, 20144.

[16] C. a. V. C. Miller, "Adventures in Automotive Networks and Control Units," in *DefCon 21*, Las Vegas, NV, USA, 2013.

[17] C. a. V. C. Miller, "A Survey of Remote Automotive Attack Surfaces," in *Blackhat USA 2014*, Las Vegas, NV, USA, 2014.

[18] F. B. A. Hartwich, "The Configuration of the CAN Bit Timing," [Online]. Available: http://www.bosch-semiconductors.de/media/ pdf_1/canliteratur/ cia99paper.pdf. [Accessed 4 June 2014].

[19] S. a. K. E. Robb, "CAN Bit Timing Requirements," [Online]. Available: http://www.freescale.com/files/microcontrollers/doc/app_note/AN1798.pdf. [Accessed 16 February 2014].

[20] CiA, "CiA 303 Part 1: Cabling and connector pin assignment," [Online]. Available: http://www.can-cia.org/index.php?eID=tx_nawsecuredl&u=2714&g=11&t=1465874134&hash=4740418813dbe7aca3d901557d01cfba19f6717f&file=fileadmin/resources/documents/groups/303_1v01080000.pdf

[21] ITS, " Status of the Dedicated Short-Range Communications Technology and Applications Report to Congress," [Online]. Available: http://www.its.dot.gov/connected_vehicle/pdf/DSRCReportCongress_FINAL_23NOV2015.pdf

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